

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
Information Technology Procurement Office (ITPO)
Justification for an Exception to Fair Opportunity
Agency-Wide Hexagon MSC Software Renewal

Description of the action being approved: This justification documents the basis for issuing a Delivery Order (DO) under the National Aeronautics and Space Administration (NASA) Solutions for Enterprise-Wide Procurement (SEWP), Government Wide Acquisition Contract (GWAC), using an exception to the Fair Opportunity process.

Description of the supplies or services required to meet the agency's needs: This competitive action establishes a Firm-Fixed-Price delivery order issued under the NASA SEWP GWAC for the Agency-Wide Hexagon MSC-software renewal. MSC's software products include integrated (such as Actran, Adams), modeling (such as Patran) and solver (such as Dytran, Nastran, Sinda) solutions and many other applications, the majority of which are in two token delivery system environments, the legacy MasterKey Plus perpetual licensed application environment and MSC One, a subscription-based system with core applications bundled into the system. The total estimated price of the proposed order is [REDACTED] and the period of performance is a one-year base period with one one-year option period commencing on April 1, 2026 with a potential end date of March 31, 2028.

Identification of Exception to Fair Opportunity:

- ☐ FAR 16.505(b)(2)(i)(A), The agency need for the supplies or services is so urgent that providing a fair opportunity would result in unacceptable delays.
- ☒ FAR 16.505(b)(2)(i)(B), Only one awardee is capable of providing the supplies or services required at the level of quality required because the supplies or services ordered are unique or highly specialized.
- ☐ FAR 16.505(b)(2)(i)(C), The order must be issued on a sole-source basis in the interest of economy and efficiency because it is a logical follow-on to an order already issued under the contract, provided that all awardees were given a fair opportunity to be considered for the original order.
- ☐ FAR 16.505(b)(2)(i)(D), It is necessary to place an order to satisfy a minimum guarantee.
- ☐ FAR 16.505(b)(2)(i)(E), For orders exceeding the simplified acquisition threshold a statute expressly authorizes or requires that the purchase be made from a specified source.
- ☐ FAR 16.505(b)(2)(i)(F), In accordance with section 1331 of Public Law 111-240 (15 U.S.C. 644(r)), contracting officers may, at their discretion, set aside orders for any of the small business concerns identified in 19.000(a)(3). When setting aside orders for

small business concerns, the specific small business program eligibility requirements identified in part 19 apply.

- ☐ FAR 16.505(b)(2)(i)(G), The order satisfies one of the exceptions permitting the use of other than full and open competition listed in 6.302 (10 U.S.C. 3406(c)(5). The public interest exception shall not be used unless Congress is notified in accordance with 10 U.S.C. 3204(a)(7).

Supporting Rationale:

ImmixTechnology is the only responsible source capable of providing the agency-wide Hexagon MSC software. MSC's software products are acquired from ImmixTechnology through the original equipment manufacturer (OEM) Hexagon. The OEM letter verified that no other vendor could provide MSC software licenses, updates, or support due to Hexagon's exclusive reseller structure. MSC's software (MSC) provides multidisciplinary simulation, modeling, and solver environment required to support NASA's mission-critical engineering programs, due to proprietary model compatibility, unique solver capabilities, and the agency's extensive legacy model base.

Market research of current simulation software shows that several major vendors—such as ANSYS, Siemens, and Dassault Systèmes—offer advanced virtual-analysis tools, yet none fully match the integrated capabilities provided by Hexagon's MSC Software. While these competing platforms focus primarily on traditional structural, thermal, and multi-physics simulation, they do not incorporate the tightly connected ecosystem that sets MSC apart.

MSC Software uniquely blends its decades-proven solvers with Hexagon's precision metrology, digital-twin technologies, and real-world sensor integration. This combination creates a closed-loop environment in which models are continuously calibrated and validated against sub-millimeter physical measurement data and high-accuracy digital-reality representations. As a result, MSC enables engineering teams—such as those at NASA—to verify spacecraft structures, materials, and mission scenarios with a level of real-world fidelity that simulation-only platforms cannot achieve.

Competitor tools provide powerful virtual analyses, but they lack MSC's ability to seamlessly integrate physical measurement and sensor data into simulations, maintain continuously updated digital twins grounded in real-world conditions, deliver real-time engineering insight through automated validation workflows, and reduce uncertainty through persistent model-to-measurement correlation. These attributes make MSC Software the only solution in its class that bridges the gap between virtual simulation and physical truth. This not only enhances reliability and mission readiness but also significantly reduces risk in environments where accuracy is critical.

MSC's simulation and CAE software technology has been used to support NASA for several years. NASA owns and maintains a variety of MSC software products including Nastran, Patran, Adams, Marc, Easy5, CAEfatigue, Actran, and Apex. These MSC software products are Commercial-Off-the-Shelf (COTS) software tools which allow users the ability to create multidisciplinary simulation models and perform structural, nonlinear, composite, dynamic,

load, thermal, aeroelastic, fatigue, and acoustic analyses. These products include multi-bodied dynamics modeling, aeroelastic simulation, 2D-3D modeling, and solver analytical solution software among other tools sets, including a unified CAE environment for virtual product development.

MSC is unique among simulation platforms because it combines decades-proven structural, thermal, and multi-physics solvers with a broader ecosystem of precision metrology, digital-twin technology, and real-world sensor integration. While competitors such as ANSYS, Siemens, and Dassault focus primarily on virtual simulation, MSC stands out by tightly linking its analysis tools to physical measurement data and high-accuracy digital-reality models. This enables NASA to validate spacecraft structures, materials, and mission scenarios with a level of fidelity that bridges the gap between simulation and real-world performance. MSC's sub-millimeter metrology, real-time engineering insight, and automation capabilities provides a uniquely closed-loop environment where models can be continuously calibrated, verified, and improved. As a result, MSC reduces uncertainty, improves reliability, and strengthens mission readiness in ways that traditional simulation-only platforms cannot match.

The Aerostructures Branch at AFRC relies on MSC software to perform the full range of structural analyses needed to support research activities and verify the airworthiness of new or modified flight hardware. These analyses—linear and nonlinear static, dynamic, vibration, and thermal, it is critical for predicting stress, deformation, buckling, and fatigue, and for establishing margins of safety required for mission success. At Glenn Research Center, the Materials and Structures and Propulsion Divisions also use MSC tools as primary FEA resources for major efforts such as Artemis, Hypersonics, Lunar Gateway, and future Mars missions. Engineers at GRC use products like Nastran, Patran, Easy5, and Adams to validate and optimize designs through virtual prototypes, supported by flexible Masterkey and MSC One licensing. LaRC similarly depends on MSC Nastran and Patran for coupled loads analysis on programs such as SLS and Commercial Crew, as well as for multi-year research efforts including SCALPSS, NGUS, ARC-LSI, SVTT, HSF, and HiCAM. LaRC also uses MSC Adams, Nastran, and Actran for multi-body dynamics, nonlinear simulations, and independent verification and validation of contractor analyses. Together, these centers demonstrate NASA's broad and sustained reliance on MSC software for high-fidelity structural modeling, mission-critical analysis, and long-term program support.

Determination that the anticipated cost to the Government will be fair and reasonable:

Pricing will be assessed against the Independent Government Cost Estimate (IGCE), historical data, previous orders, and the current commercial price list to enable the Contracting Officer to determine fair and reasonable pricing. Based on this approach, the Contracting Officer believes this approach will ensure a fair and reasonable price.

Other Facts:

None.

A statement of the actions, if any, the agency may take to remove or overcome any barriers that led to the restricted consideration before any subsequent acquisition for the supplies or services is made:

The CO will continue to examine the market in the future for alternative solutions or new sources before executing any subsequent acquisitions for the same requirements.